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Paleomagnetism of Upper Devonian reefal limestones, Canning basin, Western Australia

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ABSTRACT

Two conflicting models have been proposed for the middle Paleozoic position of Gondwana with respect to Euramerica. A pole from Morocco (Msissi Norite) that may be of Late Devonian age suggests that a broad ocean (~3,000 km) separated the northern continents from Gondwana, whereas results from Mauritania (Gneiguira Supergroup) indicate that this ocean had closed by Devonian time. Detailed thermal and alternating-field demagnetization experiments have been carried out on red and gray limestones from seven sites (89 samples) in an essentially undeformed Late Devonian reef complex, northern Canning basin, Western Australia. After minor tilt correction, the site mean characteristic directions average 42.7°/-28.9° (declination/inclination) with k = 61.6 and α_{95} = 7.8°. These results position northwestern Australia at 15°S latitude in the Late Devonian, a paleolatitude consistent with the presence of a well-developed reefal sequence. An early age is inferred for these magnetizations on the basis of the occurrence of mixed polarities and the fact that no post-Devonian directions for Australia resemble the observed declinations and inclinations. If the Smith and Hallam reconstruction for Gondwana is used, the paleomagnetic pole calculated for the Late Devonian of Western Australia agrees with the Msissi Norite result from Morocco and suggests that an ocean still separated Gondwana from Euramerica in the Late Devonian. The pole measured from the Gneiguira Supergroup in Mauritania may be based on remagnetizations of late Paleozoic age.

The directions determined from Canning basin limestones are similar in inclination to but differ by 40° in declination from paleomagnetic results from the Upper Devonian Comerong Volcanics, Lachlan fold belt, southeastern Australia. Canning basin samples are from the stable craton, and Comerong Volcanics samples are from the Tasman fold belt. The discrepancy between the two Australian results may be best explained by tectonic rotation of the Comerong Volcanics locality.

INTRODUCTION

Reliable paleomagnetic results are scarce for the middle Paleozoic of Gondwanaland (Van der Voo, 1982). This is unfortunate because it is an important time in Earth history as the supercontinent Pangea is thought to have formed in the middle or late Paleozoic. On the basis of existing paleomagnetic evidence, two, different models for the Late Devonian paleogeographic position of Gondwanaland have been proposed by Kent and others (1984) and illustrated by Scotese (1984). Both models use the Smith and Hallam (1970) reconstruction for the southern continents. Model A (Fig. 1) implies that the supercontinent Pangea had formed by Late Devonian time. The position of Gondwana in model A is based on a single paleomagnetic pole from the Early to Middle Devonian Gneiguira Supergroup in Mauritania, northwest Africa (Kent and others, 1984). In this model, the Australian continent has nearly its present-day geographic orientation, and the Canning basin (in northwest Australia) is at ~42°S paleolatitude. The position of Gondwana in Late Devonian model B (Fig. 1) is based on a single paleomagnetic pole from Morocco (Hailwood, 1974), and this model shows a broad ocean (~3,000 km) separating the northern and southern continents. In model B, Australia is rotated in a counterclockwise direction from its present orientation, and the Canning basin is at ~30°S paleolatitude. A number of workers (Kent and others, 1984; Hurley and others, 1985; Hurley and Van der Voo, 1985; Schmidt and others, 1986) have discussed the possibility that the Gneiguira Supergroup was remagnetized in the late Paleozoic. Kent and others (1984) discussed the problems of the Msissi Norite pole, including a large Holocene overprint on the natural remanent magnetization (NRM), scatter in directions after treatments of only 30 to 45 mT, and a lack of field evidence for original magnetization. In addition, results from Salmon and others (1986) suggest a Jurassic, rather than Late Devonian, age for the Msissi Norite. If this is true, then the Msissi Norite pole must be judged unreliable.

Other poles available for the Late Devonian of Gondwanaland are even less reliable (McElhinny and Valencio, 1981; Van der Voo, 1982; Kent and others, 1984) and do not provide a clear choice between the two reconstruction options. These other poles are judged unreliable because of poor age constraints (for example, Mereenie Sandstone, Embleton, 1972) or because they are from mobile belts that have problems such as rotated sub-plates or displaced terranes (Embleton, 1981). A new, apparently reliable, Late Devonian result from the Comerong Volcanics in southeastern Australia (Schmidt and others, 1986) is discussed in detail in a later section.

The purpose of this study is to determine a paleopole for Upper Devonian reefal limestones in the Canning basin, Western Australia. The differences in predicted paleolatitude and orientation of Australia make this the key continent to resolve the conflict between Late Devonian models A and B. Devonian reef complexes in the northern Canning basin are ideal for paleomagnetic studies because they are well dated and undeformed, they lie on the stable craton, and many of the rocks are ferruginous. Schmidt (1976) did reconnaissance paleomagnetic work on the Devonian reef complexes, and his results were encouraging but inconclusive because of overprints and incomplete demagnetizations.

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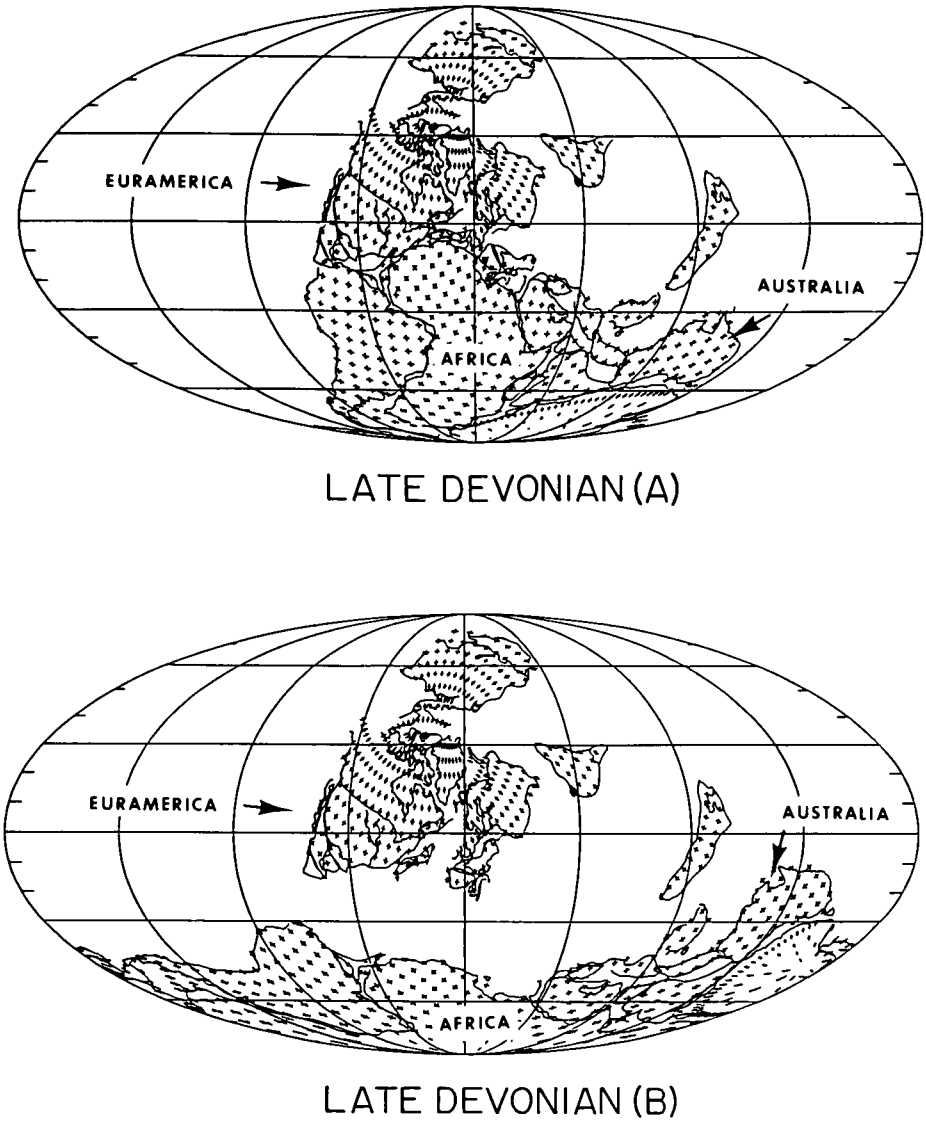


Figure 1. Alternative Late Devonian plate-tectonic reconstructions illustrated by Scotese (1984). Note differences in latitude and orientation of the Australian continent in the two models.

Considering the importance of a Late Devonian pole for Gondwana and the potential for success using Canning basin limestones, it was felt that a new paleomagnetic study was in order.

GEOLOGIC SETTING

Paleomagnetic samples were collected from a group of limestone ranges exposed along the northern margin of the large (>400,000 km^2) intracratonic Canning basin (Fig. 2). The limestone ranges are karsted, flat-topped remnants of Devonian reef complexes that rise from a few metres to >60 m in height above the surrounding alluvial plains. These ranges are essentially exhumed fringing reefs that formed along the paleoshoreline of the emergent Proterozoic Kimberley Block (Playford, 1980). Carbonate buildups of Givetian (Middle Devonian), Frasnian, and Famennian (both Late Devonian) age are exposed in the study area. Figure 3 shows a diagrammatic stratigraphic section for the reef complexes. Ages are well constrained by extensive studies of conodonts (reviewed by Nicoll, 1984), plants (Playford, 1976), and macrofossils such as stromatoporoids, brachiopods, corals, bryozoans, and fish (reviewed by Playford, 1980, 1984). The reef complexes lie on a block known as the "Lennard Shelf," and structural deformation is minor except for a few faults, numerous fractures, and slight amounts of postdepositional tilting (Playford, 1981; Hurley, 1986).

The first detailed geologic maps of the Canning basin Devonian reefs were published by Playford and Lowry (1966). Logan and Semeniuk (1976) reinterpreted the limestone ranges as structural-metamorphic products rather than ecologic or stratigraphic reefs. Playford (1980) published an updated interpretation of the Devonian reef complexes that incorporated work done since the Playford and Lowry (1966) volume, and therein he refuted the Logan and Semeniuk (1976) model. Among other observations, studies cited by Playford (1980) showed that hundreds of conodonts from the reef complexes had color-alteration indices of 1, suggesting no metamorphism in the area. Forman and Wales (1981) and Towner and Gibson (1983) published regional summaries of the geology of the Canning basin, and Purcell (1984) edited a Canning Basin Symposium volume that contains a wealth of information on the basin and on the Devonian reef complexes. Preliminary results of the paleomagnetic study herein presented were reported by Hurley and others (1985) and Hurley and Van der Voo (1985).

SAMPLE LOCALITIES

In the 1983 field season, oriented blocks were sampled from 11 localities in a reconnaissance study of magnetization in the Devonian reef complexes. Oriented blocks were collected using a magnetic compass and a Commonwealth Scientific and Industrial Research Organization (CSIRO) brass and plastic tripod-like device having level bubbles along the strike and dip directions (Embleton, 1979). Field work in 1984 concentrated on detailed sampling of localities that had the highest magnetic intensities and the most consistent magnetic directions in the reconnaissance studies of the previous year. Cores were sampled using two portable gasoline-powered drills (one of which was loaned by CSIRO), and cores were oriented using a magnetic compass and inclinometer. Mechanical problems of the drills and the presence of thin, platy bedding at some of the localities forced the collection of additional oriented blocks. Secular variation in the Earth's magnetic field was, one hopes, averaged by collecting samples from several beds over a stratigraphic interval of 1 to 3 m at each locality. A correction for the present-day declination of N3°E was applied to all of the sample orientations.

Samples from seven localities have been used to determine a paleomagnetic pole for the Devonian of the Lennard Shelf. Figure 2 shows site locations, and Figure 3 shows the stratigraphic locations of detailed sample sites, which range from the Frasnian basin (Waggon Pass, Brooking Creek, and *Frutexites* localities) and marginal-slope (McWhae Ridge stromatolite and

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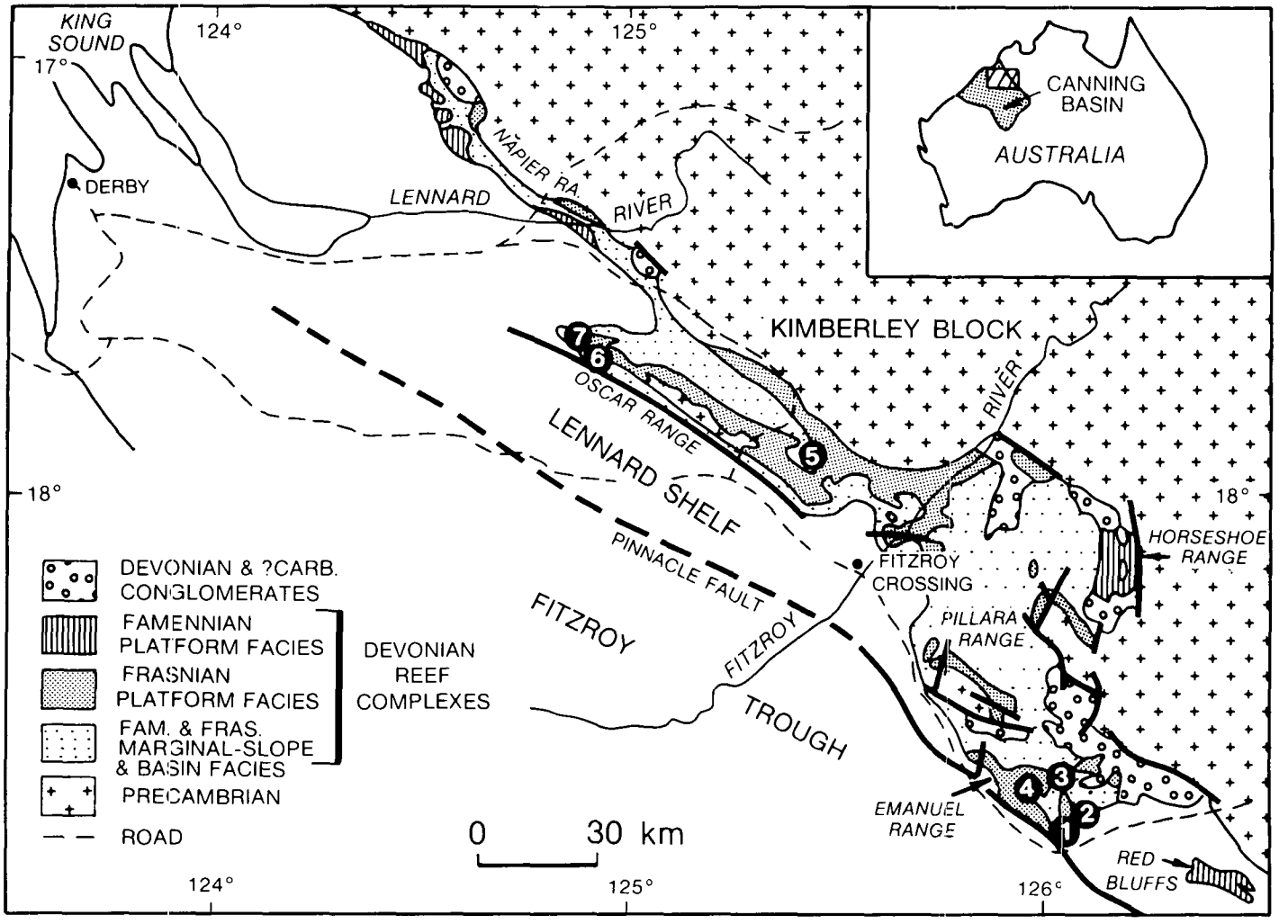


Figure 2. Geologic map of Late Devonian reef complexes, showing location of paleomagnetic sampling localities: (1) *Frutexites,* (2) McWhae Ridge stromatolite, (3) Waggon Pass, (4) Lloyd Hill, (5) Brooking Creek lamproite and limestone, (6) Palm Spring lamproite and limestone, and (7) Nullara Limestone. Diagram after Playford (1984).

Lloyd Hill localities) to the Famennian marginal-slope (adjacent to the Palm Spring intrusive) and backreef (Nullara Limestone) facies. All of the samples are limestones except the leucite lamproites at the Palm Spring intrusive (discussed in Hurley 1986; N. F. Hurley and P. W. Schmidt, unpub. data).

Nullara Limestone, *Frutexites,* McWhae Ridge stromatolite, and Waggon Pass samples are all red limestones in which hematite is the principal magnetic carrier. The Nullara Limestone is a fenestral oolitic grainstone that was deposited in a high-energy, peritidal, oxidizing environment in which hematite was probably a primary mineral. Red limestones from the other three localities formed in basinal settings that had very slow sedimentation rates. For example, the sedimentation rate at the *Frutexites* locality is estimated to be a low 1.8 mm/1,000 yr on the basis of the presence of 6 conodont subzones, with no observed hiatus, in 5.5 m of strata (Playford and others, 1984). Organic matter in open-marine sediment commonly reduces hematite to pyrite at shallow depths below the sediment/water interface. In areas of slow sedimentation, however, bacteria may oxidize organic material prior to burial so that hematite can be preserved (Wilson, 1975, p. 90). At *Frutexites* localities elsewhere in the world, the primary precipitate is thought to be an iron hydroxide (Walter and Awramik, 1979). In cases of shallow burial, however, this precursor converts to hematite. The conclusion for the *Frutexites,* McWhae Ridge stromatolite, and Waggon Pass localities is that hematite either precipitated from sea water or was recrystallized from iron hydroxide at an early stage. Other examples of red limestones in basinal settings are known in many localities, including the Devonian of Morocco (Wendt and others, 1984), the Silurian of the Michigan basin (Mesolella and others, 1974), and the Jurassic of the Bavarian Alps (Mathur, 1975).

Gray and tan limestones were sampled in marginal-slope deposits at Lloyd Hill and near the Brooking Creek and Palm Spring intrusives. With the exception of the contact-metamorphic zone at Palm Spring, magnetic intensities are moderate to low, and demagnetization studies show that hematite, goethite, and magnetite are all magnetic carriers.

Tilt corrections are minor at all localities. Numerous geopetal measurements in marginal-slope deposits at McWhae Ridge (Playford, 1980) show a 7° post-depositional steepening of dip. A 7° tilt correction was therefore applied in marginal-slope deposits at the McWhae Ridge stromatolite and *Frutexites* localities. At Lloyd Hill, geopetal measurements (P. E. Playford, 1985, personal commun.) were used to flatten

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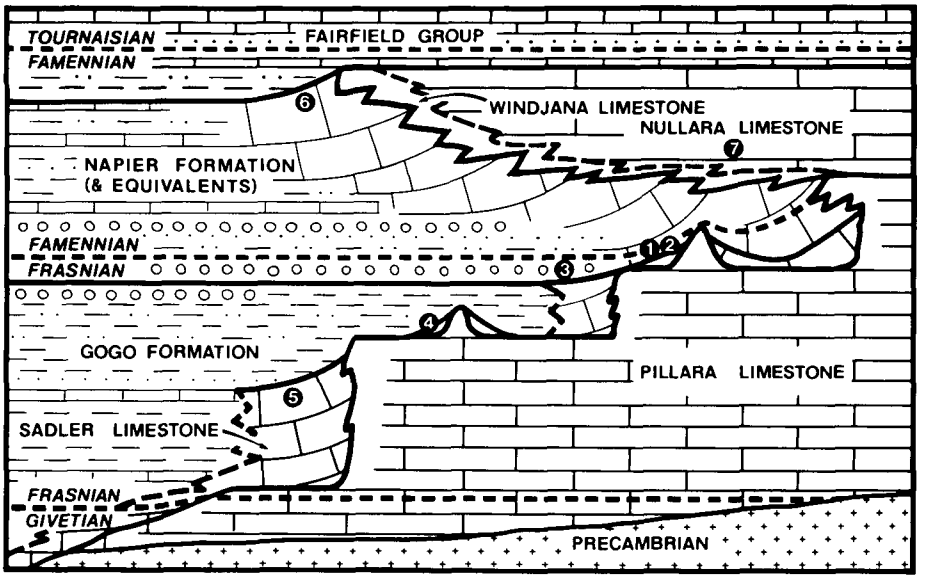


Figure 3. Stratigraphic section showing age and facies of paleomagnetic sampling localities. Diagram after Playford (1984).

orientations by 10° for samples from the northeast flank and to steepen orientations by 1.5° for samples from the southwest flank of the pinnacle reef. No geopetal measurements were available for limestones from the Palm Spring intrusive area, but a 5° tilt correction was applied by analogy with the McWhae Ridge localities. Nautiloid lineations on bedding planes in forereef deposits at Waggon Pass suggest a small amount of primary dip, and so tilt corrections of 2° to 5° were applied at this locality, leaving a 5° depositional dip. Platform deposits in the Nullara Limestone were rotated to horizontal by applying tilt corrections ranging from 7° to 10°.

LABORATORY METHODS

All oriented core samples (2.54 cm diameter) were cut into 2.2-cm-long specimens (specimen volume 10 cm^3) in the laboratory. In all of the samples, the weathered rind (top 1 cm) was first removed and discarded. Natural remanent magnetism (NRM) of each specimen was then measured using a Superconducting Technology cryogenic magnetometer; intensities commonly ranged from 10^{-2} to 10^{-4} A/m. NRM results were used to select samples for thermal and alternating-field (AF) demagnetization. A number of samples having low intensity and/or present-day directions for the NRM were excluded from further study.

Thermal demagnetizations were carried out in a Schonstedt TSD-1 apparatus or in a homebuilt non-magnetic furnace located in a field-free space (10 nT or better). AF demagnetizations were performed using a Schonstedt GSD-1 instrument capable of reaching 100 mT peak field. Data were displayed on orthogonal plots (Zijderveld, 1967) that are the projections of the end points of the magnetic intensity vectors onto the horizontal and vertical planes. Each specimen was treated with 10 to 20 thermal or AF demagnetization steps, and characteristic directions were determined by visual inspection of Zijderveld plots followed by principal-component analysis (Kirschvink, 1980) of straight-line segments made up of 4 or more points. Directions were not accepted unless mean angular deviation (MAD) angles were $<10°$, and most MAD angles were $<5°$.

PALEOMAGNETIC RESULTS

Reconnaissance work after the 1983 field season showed that a combination of thermal and AF stepwise demagnetization was the best "cleaning" technique for all of the localities, although simple thermal demagnetization was sufficient in many of the specimens. Inferred Devonian characteristic directions with northeast declination and shallow upward inclination are present in specimens from all of the localities (for example, Figs. 4a and 4b). Reversed directions with southwest declination and shallow downward inclination (for example, Figs. 4c and 4d) are common at the Lloyd Hill, *Frutexites,* Nullara Limestone, and Waggon Pass localities; reversals are not observed at other localities. Figures 5a and 5b show interpreted Devonian characteristic directions for 89 samples and site means for 7 sites. Table 1 summarizes characteristic directions observed at each of the sample localities. The site mean declination and inclination is D/I = 42.7°/–28.9° (precision parameter k = 61.6; α_{95} = 7.8°).

A high-coercivity, low–blocking-temperature (150 °C) mineral, presumably diagenetic goethite, is present in many samples and commonly carries a magnetization with a direction conforming to the present-day declination and inclination (Figs. 4a, 4c, and 4d). At localities where goethite was a problem (Waggon Pass, Lloyd Hill, and Nullara Limestone), specimens were heated and measured at 100 and 200 °C to remove the goethite effect. After this removal, AF or continued thermal demagnetization revealed a direction in most specimens with a northeast declination and a shallow upward inclination.

TABLE 1. CHARACTERISTIC DIRECTIONS OF DEVONIAN LIMESTONES

Site	PDF*	Composite†	Random§	Devonian**	N/n††	Dec./inc. (°)§§	k***	α_{95}†††
Frutexites	3	..	..	31(77%)	31/34	37.5/–31/3	22.9	5.5°
McWhae Ridge stromatolite	12	11	..	23(100%)	23/46	39.8/–38.3	25.1	6.0°
Waggon Pass	1	..	6	18(56%)	18/25	45.4/–15.3	9.4	11.9°
Lloyd Hill	2	..	7	7(43%)	7/16	30.7/–25.4	9.9	20.2°
Brooking Creek	..	..	..	3(100%)	3/3	53.4/–36.6	10.6	40.0°
Palm Spring	1	..	..	2(100%)	2/3	48.5/–23.2	37.6	41.9°
Nullara Limestone	1	..	..	5(60%)	5/6 89/133	43.9/–30.9	17.0	19.1°

*Demagnetized direction conformed to the present-day field geomagnetic direction.
†Composite diagram in which Zijderveld plot was curved; different directions were apparently carried by two mineral populations with overlapping blocking temperatures.
§Demagnetized direction was random, perhaps lighting affected.
**Characteristic direction or inferred Devonian direction. The percentage of sample directions with normal polarity shown in parentheses.
††Number of sample directions used in calculations/number of samples measured.
§§Declination/inclination.
***Precision parameter.
†††Radius of 95% circle of confidence.

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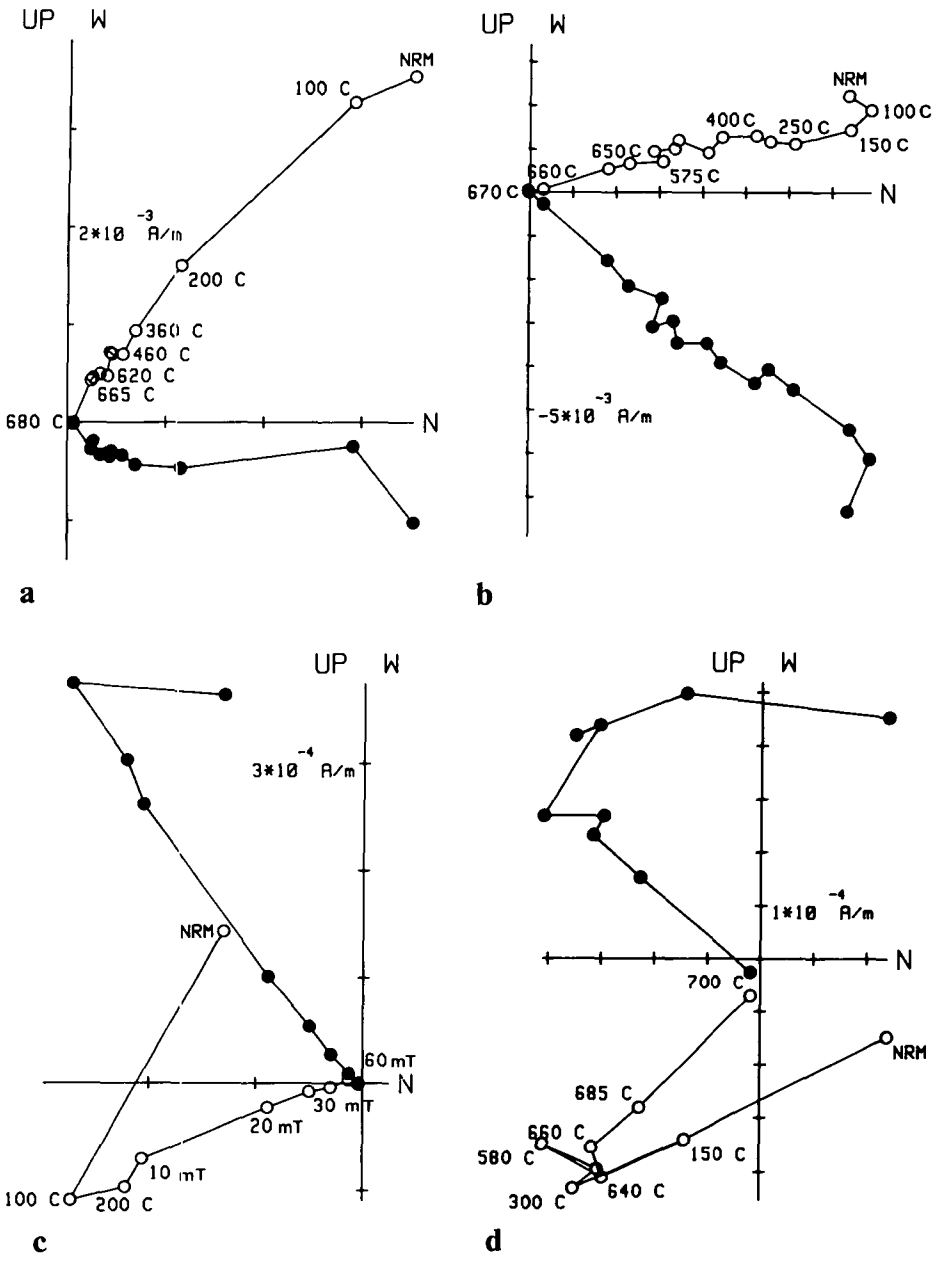


Figure 4. Orthogonal demagnetization diagrams (Zijderveld, 1967) for Devonian limestones. Coordinates are uncorrected for structural tilt. Closed circles are projections onto the horizontal plane, and open circles are projections onto the north-south vertical plane. (a) Normal-polarity Devonian result, thermal demagnetization, *Frutexites* locality, sample F2B. (b) Normal-polarity Devonian result, thermal demagnetization, Nullara Limestone locality, sample NH57C. (c) Reversed-polarity Devonian result, combined thermal and alternating-field demagnetization, Lloyd Hill locality, sample NH20A. (d) Reversed-polarity Devonian result, thermal demagnetization, *Frutexites* locality, sample F10BD.

A number of specimens had moderate-intensity random directions that could be removed by AF but not by thermal demagnetization. These random directions may have originated from lightning strikes that are common in the West Kimberley area during monsoonal rain storms (Idriess, 1949). Boulders apparently shattered by lightning have been observed at several localities in the Oscar Range. These boulders are commonly located at ridge tops far above any bush-fire heating effects, and they look freshly split because epilithic algae coat outer surfaces but not fracture surfaces. Such localities were avoided in paleomagnetic sampling. P. W. Schmidt (1984, personal commun.) believed that many of his Lennard Shelf samples (Schmidt, 1976) were lightning affected. Similar lightning problems were observed in South Africa by Graham (1961).

Most of the samples from the McWhae Ridge stromatolite locality had univectorial magnetizations and showed the typical northeast declination and shallow upward inclination observed in other Devonian samples. Several samples from this locality, however, had curved or composite Zijderveld plots that could not be resolved using AF or thermal demagnetization. Composite samples apparently have two magnetic mineral populations that have overlapping blocking temperatures. Such samples were rejected and were not used in characteristic direction determinations.

Figures 6a and 6b show directions of samples that were rejected because they had present-day and random directions, respectively. A small amount of overlap exists between distributions of present-day and characteristic directions. Specimens having directions that fall in the overlap area had both present-day and characteristic components (for example, Fig. 4a) that could be differentiated using principal-component analysis.

AGE OF THE CHARACTERISTIC MAGNETIZATION

The following lines of evidence suggest that the observed characteristic directions are Late Devonian in age.

1. The presence of normal and reversed polarities suggests, but does not prove, early magnetizations and argues against remagnetization by a subsequent short-lived metamorphic or thermal event (McElhinny, 1973, p. 85). The best locality for mixed polarities is the *Frutexites* locality. *Frutexites* were mat-forming *Cyanobacteria* (Early Proterozoic to Jurassic; Walter and Awramik, 1979) with an ability to concentrate iron, iridium, platinum, cobalt, and a number of other metals (Playford and others, 1984). Most workers suspect that *Frutexites,* perhaps in association with iron bacteria, precipitated iron hydroxides on the sea floor (Walter and Awramik, 1979), and this precursor was later altered to hematite. Early hematite and syn-depositional acquisition of the magnetic signature is suggested by the presence of layer-parallel polarity intervals (three normal and two reversed zones) in 14.5 cm of stratigraphic section at the *Frutexites* locality (Hurley and Van der Voo, 1986).

2. From the observed site mean inclination, a paleolatitude of 15°S can be calculated for Devonian reefs on the Lennard Shelf. This paleolatitude is consistent with the presence of a well-developed reef complex in the study area. If Devonian climatic belts were anything like those

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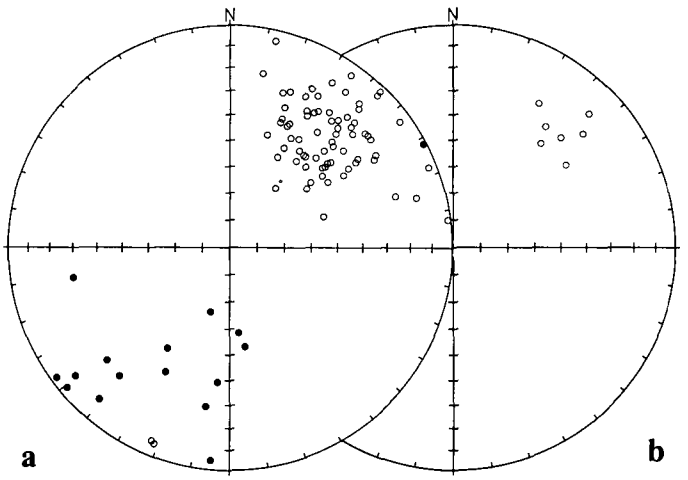


Figure 5. Equal-area stereonet projection of (a) characteristic directions of Devonian limestones after minor tilt corrections for 89 samples, sample mean = 40.1°/–29.9° (declination/inclination), k = 14.5, α_{95} = 4.1°, and (b) site means of tilt-corrected Devonian limestones for 7 sites, mean = 42.7°/–28.9° (declination/inclination), k = 61.6, α_{95} = 7.8°. Open circles are projections onto upper hemisphere; closed circles are projections onto lower hemisphere.

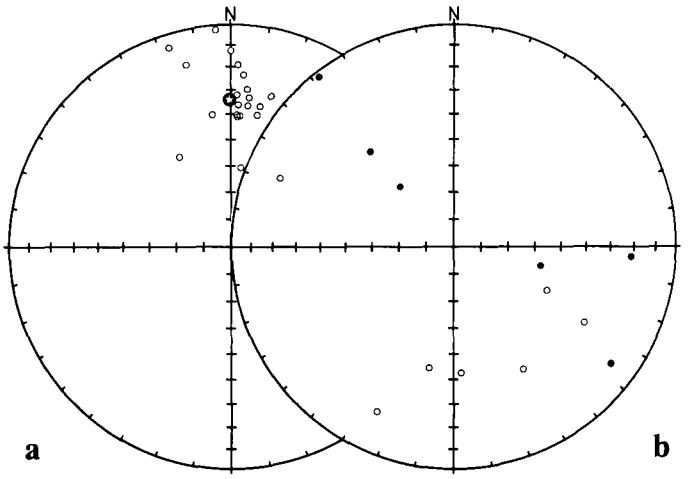


Figure 6. Equal-area stereonet projection of demagnetized directions of rejected samples that showed (a) present-day directions (20 samples) and (b) random directions (13 samples). Open circles are projections onto upper hemisphere; closed circles are projections onto lower hemisphere; star is upper hemisphere projection of geocentric axial dipole field.

of today, the northern Canning basin would be in the equatorial doldrums (Heckel and Witzke, 1979). This point is consistent with the lack of evaporites in the reef complexes because evaporites tend to form in hot, dry belts on either side of the equator. The nearest Upper Devonian evaporites are sabkha deposits in the southern Canning basin (Mellinjerrie Limestone; Howell, 1984; Lehmann, 1984), a distance of 400 km and a present-day latitude difference of 4° from the Lennard Shelf.

3. Shallow inclinations and near-equatorial paleolatitudes observed in this study could not have formed in post-Devonian time. The post-Devonian polar wander path for Australia is well known (Embleton, 1981), and paleogeographic and paleontologic data show that Australia was near the south pole by Permian time (Scotese, 1984; Nicoll, 1976). In post-Permian time, Australia drifted slowly northward to its present-day tropical and subtropical latitudes. Even though present-day and Devonian latitudes and inclinations are similar, Devonian declinations can be easily differentiated from present-day declinations because of the 40° difference between them.

4. Hematite in red backreef limestones such as the Nullara Limestone was observed to be commonly leached along hairline fractures and around very coarse fenestral pores. The cement filling these fractures and pores is a non-ferroan, homogeneous, bright-luminescing blocky calcite that is a burial cement according to petrographic and carbon and oxygen isotopic studies (Hurley, 1986). These cements post-date a late Carboniferous unconformity and pre-date an inferred Late Permian cement, evidence that dates hematite as pre-Permian.

To further establish a Devonian age for magnetization, fold tests (McElhinny, 1973) were attempted by sampling the northeast and southwest flanks of the Lloyd Hill pinnacle reef and the limbs of a gentle compactional fold in the Nullara Limestone. The concept in regard to Lloyd Hill was that a small increase of dip, perhaps a 5° addition to depositional dip on each flank, may have occurred with burial and compaction. Unfortunately, neither fold test was conclusive because of the slight amount of tilting involved.

A NEW DEVONIAN PALEOPOLE: DISCUSSION

The observed Devonian directions (declination/inclination = 42.7°/–28.9°) and shallow paleolatitude (15°S) calculated for the Lennard Shelf favor Late Devonian model B over model A (Fig. 1). This implies that a broad ocean separated the northern and southern continents and effectively dates the formation of the supercontinent Pangea as post-Devonian.

Figure 7 shows the middle to late Paleozoic apparent polar wander path (APWP) for the Gondwana continents using the Smith and Hallam (1970) plate reconstruction. All poles used in making this diagram have been rotated to African coordinates (Table 2). The most important poles are labeled "CB" for the Canning basin, "MN" for the Msissi Norite in Morocco (Hailwood, 1974), and "GN" for the Gneiguira Supergroup in Mauritania (Kent and others, 1984). Pole MN was used to make Late Devonian model B (Fig. 1). Even though the age of pole MN is now in doubt (Salmon and others, 1986), the plate reconstruction that used this pole may be accurate based upon the agreement between the MN pole and the new, apparently

TABLE 2. PALEOPOLE CALCULATIONS, DEVONIAN LIMESTONES, CANNING BASIN, WESTERN AUSTRALIA

Present-day latitude, longitude:	18°S, 125°E
Site mean declination, inclination:	42.7°, –28.9°
Site mean precision parameter, k:	61.6
Site mean α_{95}, seven sites:	7.8°
South pole position:	49.1°S, 38.0°E

Gondwana model	Rotation	Rotation parameters Lat.	Long.	G	Calculated paleopole: African coordinates latitude, longitude
Smith and Hallam (1970)	Australia-Antarctica	–3.6°	+40.0°	–31.0°	40.0°S, 9.7°E
	Antarctica-Africa	+1.3°	–36.0°	+58.4°	7.8°N, 22.5°E*

*Plotted as paleopole CB, Figure 7.

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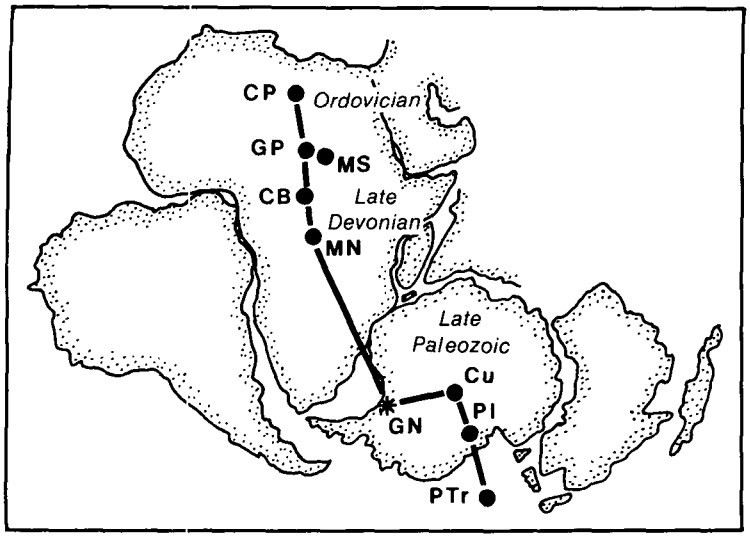


Figure 7. Middle through late Paleozoic apparent polar wander path for Gondwana, using Smith and Hallam (1970) reconstruction. Paleopoles are CP = Chapmans Peak, South Africa (Hurley and others, 1985); GP = Gydo Pass, South Africa (Hurley and others, 1985); CB = Canning basin, Australia (this study); MN = Msissi Norite (Hailwood, 1974); MS = Mereenie Sandstone (Embleton, 1972); GN = Gneiguira Supergroup (Kent and others, 1984). Cu, Pl, and PTr are the well-established late Paleozoic mean poles for Gondwana.

reliable CB pole of this study. Pole GN falls in an area of Carboniferous and Permian poles, and this evidence suggests that Devonian rocks from the Gneiguira Supergroup (pole GN) were remagnetized in the late Paleozoic.

It must be noted that the pre-Devonian APWP segment of Figure 7 is uncertain and is shown only to illustrate the proximity of our Late Devonian CB pole to the Ordovician CP pole (Hurley and others, 1985), as well as to other Ordovician Gondwana poles. Hargraves and Dawson (1985), however, have reported preliminary results for ring complexes in Niger, which on the basis of a Silurian age (R. B. Hargraves, 1986, personal commun.), would require an excursion of the APWP between the Ordovician and Devonian poles.

Three major problems must be addressed before either Late Devonian model of Kent and others (1984) can be conclusively favored. These are (1) Does the Smith and Hallam (1970) reconstruction represent the best fit for the Gondwana continents? (2) Are there alternate positions for Euramerica in the Late Devonian? (3) How do paleomagnetic results compare to paleoclimatic and paleontologic data for the Late Devonian?

Smith and Hallam Reconstruction

Embleton and others (1980) and, more recently, Lawver and Scotese (1985) have proposed a modification to the long-accepted Smith and Hallam (1970) computer/geologic fit for the Gondwana continent. The main difference in the new fit is that the Antarctic Peninsula is west rather than east of the southern tip of South America. This requires a non-rigid Antarctic plate to avoid overlap between the Antarctic Peninsula and the Falkland Plateau. Lawver and Scotese (1985, p. 59) cited "recent paleomagnetic evidence and geologic trends" in west Antarctica to support their model. Australia was joined to Antarctica, and so rotation of Antarctica to fit the new model causes a slight rotation of Australia. In the alternative fit, the position of the Canning basin pole (CB) changes only slightly. The CB result remains in agreement with the Msissi Norite (MN) pole and still suggests late Paleozoic remagnetization of the Gneiguira Supergroup pole of Kent and others (1984).

Position of Euramerica

The position of Euramerica shown in Late Devonian models A and B is based largely on results from the Catskill delta in the Appalachians (Kent and Opdyke, 1978; Van der Voo and others, 1979). New Catskill results (Miller and Kent, 1985) suggest a syn-folding (Carboniferous?) origin for the magnetization and cast doubt on the Devonian age of earlier poles. Early Devonian results from the Canadian Arctic (Dankers, 1982) place Euramerica ~20° farther south than previous models allowed and suggest that the position of Euramerica could have been farther south than is shown in Late Devonian models A and B.

Paleoclimatic and Paleontologic Evidence

Caputo and Crowell (1985) have shown that Late Devonian glacial deposits are present in Brazil and parts of Africa. The proximity of the CB pole to these deposits provides paleoclimatic support for the south pole position inferred from Canning basin paleomagnetic studies.

Several Devonian paleogeographic reconstructions have been proposed on the basis of lithofacies distributions, biotic distributions, and paleoclimatic models similar to present-day models (Boucot and Gray, 1979; Heckel and Witzke, 1979; Ziegler and others, 1979; Scotese and Barrett, 1985). Ziegler and others (1979) favored a model in which the northern and southern continents are separated by a broad ocean. Models of Heckel and Witzke (1979) and Scotese and Barrett (1985) favored an assembly that approximates Pangea at the end of Devonian time. Heckel and Witzke (1979) achieved this assembly by assigning a more southerly latitude to Euramerica, whereas Scotese and Barrett (1985) preferred a more northerly position for Gondwana. Boucot and Gray (1979) hypothesized a Pangean assembly throughout the Paleozoic.

Paleontologic data from a number of different fossil groups indicate a high degree of endemism in the Early Devonian, followed by increasingly cosmopolitan populations at the end of the Devonian. This relationship is supported by studies of fish (Young, 1981, 1985), conodonts (Charpentier, 1984), plant microfossils (Playford, 1976; Truswell, 1981), and brachiopods (Boucot and others, 1969). All of these data suggest that a barrier to migration existed between the northern and southern continents through Middle Devonian time. This barrier apparently broke down in the Late Devonian and allowed interchange between Gondwana and Euramerica.

Unfortunately, migration mechanisms are poorly understood. Larval stages of marine organisms could disperse along warm equatorial currents, fish could swim, and spores and pollen could be carried great distances by the wind. The unresolved question is: How broad is too broad when it comes to migration across a Paleozoic ocean? Paleogeographic and paleontologic data allow us to conclude only that the broad sea between the northern and southern continents narrowed through Devonian time. Quantitative estimates of the widths of ancient oceans must be obtained from paleomagnetic studies. From the present study (and an anticipated revision of the Catskill pole), we infer that a first-land or marine-shelf connection may have been established between North and South America during the Late Devonian, although a

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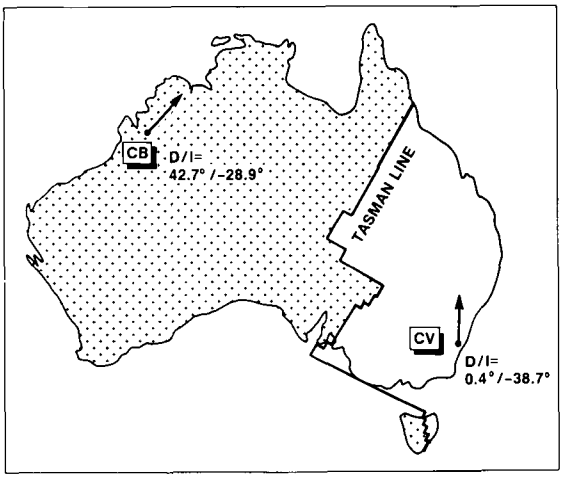


Figure 8. Comparison of Late Devonian characteristic directions determined from Canning basin limestones (CB) and Comerong Volcanics (CV). The Tasman Line (after Powell, 1984) marks the eastern edge of proven Precambrian basement in Australia. D/I = declination/inclination.

wide ocean may still have separated Africa and Europe.

COMPARISON OF CANNING BASIN AND COMERONG VOLCANICS RESULTS

Schmidt and others (1986) have reported a paleopole from the Upper Devonian Comerong Volcanics in southeastern Australia. These volcanic rocks are well dated and are essentially identical in age (Frasnian, ~370 Ma) to Canning basin limestones of this study. Measured inclinations in the Canning basin limestones and Comerong Volcanics are remarkably similar (-28.9° and -38.7°, respectively), given today's latitudinal separation between the two localities. A 40° difference in declination is present, however, between the two results (Fig. 8). Because the two localities are on the same continent, the large declination discrepancy must be explained either by remagnetization at one locality or by tectonic rotation between localities.

A positive fold test argues against remagnetization of the Comerong Volcanics at any time later than early Carboniferous. Evidence suggests that Canning basin limestones carry a depositional remanent magnetization (DRM), although Schmidt and others (1986) suggested the possibility of an early Carboniferous chemical remanent magnetization (CRM). The presence of mixed polarities at most localities and the presence of discrete, layer-parallel magnetostratigraphic intervals at the *Frutexites* locality (Hurley and Van der Voo, 1986) suggest that the Canning basin magnetizations are very early. We note, moreover, that an early Carboniferous age of magnetization for the Canning basin pole would complicate the story of the assembly of Pangea to a large extent. The CB pole of Figure 7 would, if it is of early Carboniferous age, imply that a wide ocean existed at that time. Given that the Hercynian collision between Gondwana and Euramerica is generally thought to have commenced in Visean (late early Carboniferous) time, an impossibly high velocity would be needed for Gondwana.

The Canning basin is an intracratonic basin, and all sample sites are on the Lennard Shelf structural block, a part of the stable Australian craton. The Comerong Volcanics are part of the Tasman geosyncline and are in a fold belt for which cratonic coherence cannot be assumed. Because of this fact, we feel that the most likely explanation for the declination discrepancy is a Late Devonian to early Carboniferous tectonic rotation between the Canning basin and Comerong Volcanics localities.

Using tectonic grain in nearby Ordovician turbidites, Schmidt and others (1986) considered a 15° clockwise rotation of the Budawong synclinorium north of the Batesmans Bay megakink band. A clockwise rotation is used because the mean of fold axial trends in the area is 015°, and the tectonic grain north of the Batesmans Bay megakink is 030° (Powell, 1984). The paleopole calculated after removal of this rotation is in even greater disagreement with the Canning basin paleopole, and the rotation is discounted by Schmidt and others (1986). A likely possibility that was not considered, however, is a 15° counterclockwise rotation of the Budawong synclinorium south of the Batesmans Bay megakink band. There is no reason to assume that the mean of fold axial trends (015°) is the line about which rotations occurred. The paleopole resulting from removal of the counterclockwise rotation is in better agreement with the Canning basin pole.

CONCLUSIONS

Detailed thermal and alternating-field demagnetization experiments have been carried out on ferruginous limestones from seven sites in the northern part of the intracratonic Canning basin, Western Australia. All sites are in well-dated, essentially undeformed Late Devonian reef complexes that lie on the stable craton. After minor tilt correction, the characteristic directions for 89 samples average 42.7°/-28.9° (declination/inclination) with the precision parameter k = 61.6 and α_{95} = 7.8°. These results place the reef complexes at a paleolatitude of 15°S during the Late Devonian, a position consistent with the presence of a well-developed, evaporite-free reefal sequence. Evidence for a Devonian age of magnetization includes the presence of normal and reversed polarity specimens and the fact that the observed northeast declinations and shallow inclinations could not have formed in post-Devonian time.

The new Canning basin paleopole falls in central Africa if the Smith and Hallam (1970) Gondwana reconstruction is used. The Msissi Norite (Hailwood, 1974) and Gneiguira Supergroup (Kent and others, 1984) poles have been used in Late Devonian plate tectonic reconstructions, and both poles have their problems. In the light of data presented herein, it appears that the Msissi Norite result, perhaps by coincidence, is valid for the Late Devonian, whereas the Gneiguira Supergroup was remagnetized in the late Paleozoic. New paleomagnetic evidence (Miller and Kent, 1985; Dankers, 1982) suggests that Euramerica was farther south than previously thought in either Late Devonian models A or B. Apparently, the best model to accommodate paleontologic, paleogeographic, and new paleomagnetic evidence would show a broad ocean existing between the northern and southern continents in Early Devonian time. This ocean gradually closed through Middle and Late Devonian time, but the collision that formed Pangea was apparently a post-Devonian event.

A discrepancy in characteristic directions exists between Late Devonian paleomagnetic results from the Canning basin and from the Comerong Volcanics in southeast Australia. Evidence for early magnetization is good at both localities. Owing to cratonic coherence at the Canning basin site and to the fact that the Comerong Volcanics are part of the Tasman fold belt, the discrepancy in declination is best explained by tectonic rotation of the Comerong Volcanics locality.

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